

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Define SOAP.

A.SOAP (Simple Object Access Protocol) is an XML-based messaging protocol used for communication between applications and web services.

2. Define WSDL.

A.WSDL (Web Services Description Language) is an XML-based language that describes a web service, including its operations, messages, protocols, and endpoints.

3. What is an API?

A.An API (Application Programming Interface) is a set of rules and interfaces that allows different software applications to communicate and exchange data.

4. Define XML Schema.

A.XML Schema (XSD) is used to define the structure, elements, attributes, and data types allowed in an XML document.

5. What is a Web Service?

A.A **Web Service** is a software system that enables communication and data exchange between applications over a network using standard web technologies.

6. Define REST.

A.REST (Representational State Transfer) is an architectural style for designing web services that use resources, standard HTTP methods, and commonly JSON data.

7. What is a Protocol?

A.A **protocol** is a set of rules and standards that defines how data is transmitted and communicated between devices or systems.

8. Define Endpoint in web services.

A.An **endpoint** is the specific URL or network address where a web service or API can be accessed.

9. What is a Service in web applications?

A. A **service** is a software component that performs a specific business function or provides functionality to other applications or components.

10. What is the difference between SOAP and REST?

A. SOAP is a **protocol** with a strict XML-based message format, whereas REST is an **architectural style** that commonly uses HTTP and JSON. REST is generally simpler and more lightweight, while SOAP provides stricter standards and enterprise features.

11. What is the role of WSDL in web services?

A. WSDL acts as a **service contract**. It describes the available operations, message formats, data types, communication protocols, and service endpoints.

12. What is JSON and how is it used in APIs?

A. JSON (JavaScript Object Notation) is a lightweight text-based format for representing structured data. APIs commonly use JSON to send requests and responses between clients and servers.

13. What is HTTP and why is it used in web services?

A. HTTP (Hypertext Transfer Protocol) is a communication protocol used to transfer data between clients and servers. It is widely used in web services because it is standardized, widely supported, and works well with the web infrastructure.

14. What is meant by request and response in web communication?

A. A **request** is a message sent by a client asking a server to perform an action or provide data. A **response** is the message returned by the server containing the requested result or an error.

15. What is the importance of XML in web services?

A. XML provides a structured and platform-independent format for representing and exchanging data. It is especially important in SOAP, WSDL, and XML Schema-based web services.

16. What is the purpose of an API endpoint?

A. An API endpoint provides a specific access point through which a client can request a particular resource or perform a specific operation.

17. What is the difference between HTTP and HTTPS?

A. HTTP transfers data without transport encryption, while HTTPS uses TLS to encrypt communication and provide stronger protection against interception and modification.

18. What is meant by RESTful services?

A. **RESTful services** are web services designed according to REST principles. They use resource-based URLs, standard HTTP methods such as GET, POST, PUT, and DELETE, and are typically stateless.

19. What is a namespace in XML?

A. An **XML namespace** is a unique identifier, usually expressed as a URI, used to distinguish XML elements and attributes and prevent naming conflicts.

20. What is the role of a client and server in web services?

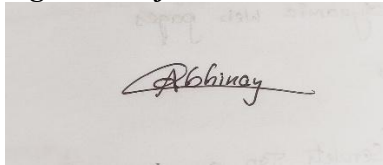
A. The **client** sends requests for data or services, while the **server** receives and processes those requests and sends back appropriate responses.

Code of Conduct:

I G. Abhinay 192373062 that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a piece of paper with a handwritten signature in black ink. The signature appears to be 'Abhinay' written in a cursive style.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively